

Education

McGill University Montreal, Canada
Vanier Scholar, Ph.D. Y4, Biological and Biomedical Engineering Sept 2022-2026

McGill University Montreal, Canada
Master's (Thesis). Biological and Biomedical Engineering, **GPA: 3.88/4.0** Sept 2020-August 2022

National Institute Of Technology (NIT) Warangal, India
Bachelor of Technology, Chemical Engineering, **GPA: 8.97/10.0** (distinction) May 2016- 2020
Research experience at Nanyang Technological University (Singapore) and Université de Montréal (Mitacs Globalink)

Work and Venture Experience

Venture Fellow Toronto, Canada
Halo Health September 2025 – Present

- Evaluated 50+ early-stage healthtech ventures across biopharma, digital health, and medical devices, assessing technical feasibility, clinical evidence, regulatory risk, and commercial viability to inform investment decisions.
- Synthesized clinical perspectives from physician-investors and translated them into actionable diligence findings for the deal team, contributing to investment recommendations.
- Sourced and screened new deal flow by mapping emerging health tech opportunities across drug delivery, digital health, and software as a medical device.

Activator Lead Montréal, Canada
Nucleate Canada September 2025 – Present

- Sourced and screened 20+ early-stage biotech startups for Nucleate's flagship Activator program, evaluating technical readiness, addressable market size, and regulatory pathway to inform cohort selection.
- Serving as primary liaison to a finalist startup, connecting the team with industry mentors and experts to advance their go-to-market strategy.

Founding Engineer Montréal, Canada
Beta Biomed Inc. May 2022 – Present

- Joined as founding R&D engineer at a McGill spin-off developing an AI-assisted antimicrobial resisting testing and nucleic acid testing platform, supported by an NSERC-CREATE MedTech fellowship.
- Led grant writing and technical narrative development that secured \$300k+ in non-dilutive funding from government and private institutions, enabling prototype development and pre-clinical testing.
- Built and iterated on a functional prototype used in clinical engagement and investor pitches, translating technical information into value propositions for clinical and industry audience.
- Conducted 15+ clinical discovery interviews with urologists and general physicians to validate unmet clinical needs and inform regulatory and reimbursement strategy.

Quebec scientific entrepreneurship program Montréal, Canada
September 2022 – Jan 2023

- Completed an intensive entrepreneurship training program focused on commercialization of scientific research, including market validation, business model development, and technology transfer.

Research Experience

McGill University Montreal, Canada
Ph.D. Researcher September 2022 – 2026

- Designing and developing wearable plasmonic biosensors for real-time, non-invasive monitoring of inflammatory biomarkers in sweat and interstitial fluid, with applications in chronic disease management.
- Contributed to securing \$500k+ in research funding by writing grant proposals and generating technical data.
- Coordinated with 20+ collaborators for a multi-year project spanning nanofabrication, surface chemistry, fluidic integration, and machine learning. This resulted in 12 peer-reviewed publications including Nature Nanotechnology and filed a US utility patent and Canadian patent application for a microfluidic detection system developed during the research program.

McGill University Montreal, Canada
Master's thesis September 2020 – August 2022

- Developed a 3D-printed microfluidic platform for rapid colorimetric detection of pathogens, integrating additive manufacturing with nucleic acid sensing to enable sample-to-answer diagnostics.

- Published first-author research in Lab on a Chip and co-authored a review in Sensors & Diagnostics synthesizing the landscape of microfluidic-based pathogen detection methods.

Patents

- Yedire, S.G.**, Isaac Hosseini, I., AbdelFatah, T., Jalali, M., Mahshid, S., Microfluidic Detection System with Adjustable Flow Control. US Utility patent App. US20240253033A, Canadian Patent App. CA3219654A1.

Publications (Selected, [Full list available here](#))

- Jalali, M., AbdelFatah, T., del Real Mata, C., Isaac Hosseini, I., **Yedire, S.G.**, McKay, Geoffrey., Corsini, R., Siavash Moakhar, R., Shieh, H., Resznetnik, Grace., Hamidi, S.V., Yansouni, C., Nguyen, D., Mahshid, S., “Ultra-rapid nanoplasmonic colorimetry in microfluidics for antimicrobial susceptibility testing directly from specimens”. **Nature Nanotechnology**, 2026.
- Mahshid, S.S., **Yedire, S.G.**, Moakhar, R.S. et al. “A nucleic acid detection device for rapid multiplexed molecular disease diagnostics”. **Nature Reviews Bioengineering**, 2025.
- Yedire, S.G.**, Isaac Hosseini, I., Siavash Moakhar, R., Mahshid, S., “Additive manufacturing leveraged microfluidic setup for sample to answer colorimetric detection of pathogen.” Lab on a Chip, 2023, 23, 4134-4145.
- Yedire, S.G.**, Khan, H., AbdelFatah, T., Mahshid, S., “Microfluidic-based Colorimetric Nucleic Acid Detection of Pathogens,” **Sensors & Diagnostics**, 2023, 2, 763-780.
- AbdelFatah, T., Jalali, M., **Yedire, S.G.**, Isaac Hosseini, I., del Real Mata, C., Khan, H., Vahid Hamidi, S., Jeanne, O., Siavash Moakhar, R., McLean, M., Patel, Dhanesh., Wang, Z., Yousefi, M., Vidal, S., Liang, C., Mahshid, S., “Nanoplasmonic Amplification in Microfluidics Enables Accelerated Colorimetric Quantification of Nucleic Acid Biomarkers from Pathogens.” **Nature Nanotechnology**, 2023, 18, 922–932.

Honors and Awards

- Vanier Canada Graduate Scholarship** | \$150,000
Canada's most prestigious doctoral award. One of four biomedical engineers nationally selected in the 2024 cohort.
- McGill Innovation Fund (MIF)** | \$100,000
Competitive commercialization grant awarded to advance technology transfer from the lab to market.
- NSERC-CREATE MedTech Talent Accelerator Fellowship**
Fellowship supporting graduate researchers working at the intersection of academic R&D and medtech startup development.
- TechAccelR Grant**
Commercialization grant including business development training for translating lab research into viable products.
- Graduate Research Enhancement and Travel Award (GREAT) to present at the Gordon Research Conference on Bioanalytical Sensors.
- McGill Engineering Doctoral Award (MEDA), funding Ph.D. research for up to 3 years.
- Graduate Excellence Fellowship (GEF) from the Department of Bioengineering.
- Mitacs Graduate and Undergraduate Fellowships for research at Canadian universities.
- 100% tuition waiver for academic excellence across all four undergraduate years at NIT Warangal, India.

Leadership Activities

Biological and Biomedical Engineering Student Society

Vice President of Academics

September 2022- 2023

- Fostered and promoted academic outreach among graduate students in the Bioengineering department at McGill University.
- Led organization of departmental research symposium bridging student research with industry stakeholders.

McGill Engine Centre

Montréal, Canada

Student Advisory Board

September 2022- 2023

- Advised McGill's Engine Centre on programming strategy to strengthen the pipeline from lab research to startup formation, contributing to enhancing student engagement with TechAccelR program.